

Jaden Solorzano

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Education

Texas A&M University, BS in Mechatronics Engineering Technology, GPA: 3.3
Minor in Embedded Systems, Greater Texas Foundation Aggie Scholar

Expected December 2026

Experience

Research Technician: Center for Infrastructure Renewal (CIR), Dept. of Civil & Environmental Engineering, Texas A&M (PI: Dr. Chenglin Wu)

July 2026 – Present

- Brought on to get an autonomous pipeline leak detection system running on a humanoid RobiDuo dual arm mobile manipulator (ROS Noetic, LiDAR SLAM, RGB-D and IR cameras). Got full open loop control of both 6-DOF RealMan arms, grippers, and the mobile base all running at the same time, using MoveJ, MoveP and MoveL to run continuous sweeping scans for gas sensor data. Modeled a 1:1 URDF in Gazebo and RViz.
- Working in Gazebo simulation towards autonomy so the robot can eventually patrol a pipeline on its own, avoid obstacles, and spot corrosion with a trained model, logging where and what it finds for a person to inspect.
- Designing and building a custom 4-inch pipe inspection crawler with adjustable dual treads and a PCB, factoring in pipe geometry, mechanical limits, power draw, and sealing to handle verticality, reducers, and T joints.

Undergraduate Researcher: Controls and Mechatronics Research Lab, Dept. of Mechatronics Engineering Technology, Texas A&M (PI: Dr. Xingyong Song)

May 2026 – Present

- Rebuilt a modular package delivery mechanism for an autonomous vehicle platform, coordinating 8 leadscrew driven linear actuators and 4 servos. The leadscrews eliminate the need for a gantry and this lets the system reconfigure at loading to fit various box sizes, stack boxes, and push different size boxes off for delivery.

Projects

Custom 3-DOF Robotic Arm with Motion Planning

April 2026

- Built a 3-DOF robotic arm from scratch with no kit, covering the mechanical design, power electronics, and embedded control; derived the forward kinematics with screw theory and Product of Exponentials, and solved the inverse kinematics with a Jacobian Newton-Raphson solver that converged within 1 mm in under 50 steps.
- Wrote three motion primitives from first principles: joint space (MoveJ), linear Cartesian (MoveL), and cubic spline interpolation for curved paths (MoveC). The cubic spline expanded a 16 waypoint circle into roughly 270 interpolated points to turn a jagged path into a smooth one, and replaced the controller's speed limiting with an EMA filter to eliminate acceleration jitter. Verified by drawing six shapes on paper with and without splines.

NASA VITAL Project Embedded Sensor Core and Telemetry Design (Capstone)

January 2026 – Present

- Sensors lead on a \$1.2M NASA funded coastal monitoring platform spanning several capstone teams, designing a modular embedded sensor core with a biodegradable enclosure, low power operation, and wireless telemetry.
 - Picked the sensors for both deployables (a floating micro-buoy and a fixed ground stake) and wrote the baseline sensor list that sets the requirements the mechanical teams design around.
 - Designed the sensor core PCB in KiCad and developing STM32 firmware in STM32CubeIDE, with core sensors on the main board and the rest on swappable packs behind a Universal Sensor Port. This allows us to deploy packs for different applications and pull packs for recalibration and maintenance.

Industrial Manipulator Programming (Universal Robots UR3e)

March 2026

- Programmed palletizing/depalletizing, stacking/destacking sequences and continuous path trajectories on an industrial UR3e 6-DOF arm and gripper, via the native controller and Python scripting over ROS2 Humble.

Additional: Autonomous SCUTTLE robot that finds and collects golf balls with HSV computer vision, closed loop control, and LiDAR obstacle avoidance (April 2025); PID light tracker with photoresistors (March 2026); ARM assembly bomb defusal game run off raw GPIO registers on an STM32F401RE with IR sensors, LEDs, and a LCD display (Nov 2024); multi stage analog guitar pedal circuit with an op amp square wave oscillator, RC filter, and clipping stage (Dec 2024); double pendulum MATLAB sim validated against a 3D printed build (Dec 2024).

Skills

Programming Languages: Python, C, C++, Assembly

Robotics & Controls: ROS1 Noetic, ROS2 Humble, Gazebo, RViz, URDF/Xacro, SLAM, motion planning, forward/inverse kinematics, control systems, state machines, sensor integration, YOLOv8, OpenCV

Design & Fabrication: Fusion 360, SolidWorks, KiCad, PCB layout, analog and digital circuits, 3D printing

Embedded & Tools: STM32 (CubeIDE), ESP32, Raspberry Pi 4, Pico (RP2040), FreeRTOS, bare metal, I2C, SPI, UART, ADC, PWM, GPIO/register level, interrupts, timers, oscilloscope, soldering, MATLAB/Simulink, Linux, Git